PROBLEM :

A signal sampled at 48 kHz is decimated by 6, but the anti-aliasing filter cutoff frequency is set to 20 kHz. Identify the design error and calculate the correct cutoff frequency

GIVEN DATA :

Parameter	Value
Original Sampling Rate (Fs)	**48 kHz**
Decimation Factor (M)	**6**
Given Anti-Aliasing Filter Cutoff Frequency (Fc)	**20 kHz**

Calculate the New Sampling Rate

Calculate the New sampling Rate

* The sampling rate after decimation is

$$F_s(\text{New}) = \frac{F_s}{M}$$

By substituting the values

$$F_s(\text{New}) = \frac{48}{6}$$

$$= 8\ \text{KHz}$$

Explanation:

After decimation, the signal is sampled 8,000 times per second instead of 48,000 times per second.

Calculate the New Nyquist Frequency

Calculate the New Nyquist Frequency

$$F_N = \frac{F_s(\text{New})}{2}$$

$$= \frac{8}{2} = 4 \text{ KHz}$$

Explanation:

After decimation, the highest frequency that can be represented without aliasing is 4 kHz.

What Changed?

We changed only one parameter:

Before	After
Cutoff Frequency = **20 kHz**	Cutoff Frequency = **4 kHz**

Why Does This Fix the Error?

With a 4 kHz cutoff:

- Frequencies below 4 kHz pass through.
- Frequencies above 4 kHz are removed.
- No high-frequency components remain to fold back after decimation.
- Therefore, aliasing is prevented.